Chain rule + small increments formula.

Small increments in 1-variable setting

Consider $f(x)$ with derivative $f'(x)$

we had

$$f(x+\Delta x) \approx f(x) + \Delta x f'(x)$$

small increments formula / linear approximation to $f(x+\Delta x)$.

Also be expressed.

$$\Delta f = f(x+\Delta x) - f(x) \approx \Delta x f'(x)$$

change = new − old

This will generalise to the multi-variable setting (2-variable) $f(x,y)$

$$\Delta f = f(x+\Delta x, y+\Delta y) - f(x,y)$$

$$\approx \Delta x \frac{\partial f}{\partial x} + \Delta y \frac{\partial f}{\partial y}$$

where both partial derivatives are evaluated at the old values (x,y)

And for a function of n-variables

$f(x_1, x_2, \ldots, x_n)$

$$\Delta f = f(\ldots, x_i + \Delta x_i, \ldots) - f(x_1, x_2, \ldots, x_n)$$

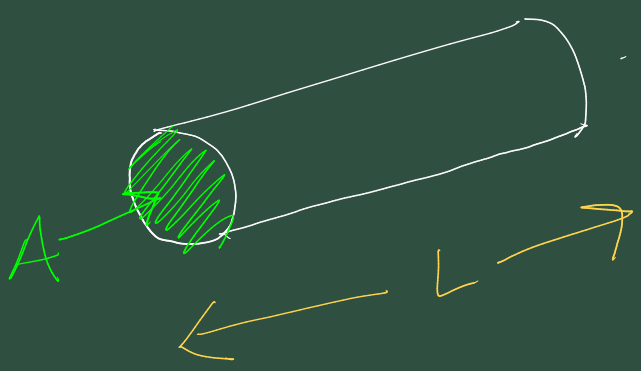
$$\approx \sum_{i=1}^{n} \Delta x_i \frac{\partial f}{\partial x_i}$$

Example The resistance R of a length L wire is given by

$$R = \rho \frac{L}{A}$$

"rho" ρ.

where L = length of the wire, A cross-sectional area, and ρ is the resistivity of the wire's material.



Suppose A decreases by 1% and L increases by 2.5%. Use the small increments formula to estimate the change in R.

$$\Delta R \approx \Delta A \frac{\partial R}{\partial A} + \Delta L \frac{\partial R}{\partial L}$$

where $\Delta A = -0.01A$

$\Delta L = 0.025L$

$$\frac{\partial R}{\partial A} = \frac{\partial}{\partial A}\left(\rho \frac{L}{A}\right) = -\rho \frac{L}{A^2}$$

$$\frac{\partial R}{\partial L} = \frac{\partial}{\partial L}\left(\rho \frac{L}{A}\right) = \rho \frac{1}{A}$$

$$\Delta R \approx -0.01A\left(-\rho \frac{L}{A^2}\right) + 0.025L\rho \frac{1}{A}.$$

$$= 0.01\rho \frac{L}{A} + 0.025\rho \frac{L}{A}$$

$$= 0.035R.$$

So the resulting change ΔR, in R, is approximately a 3.5% increase in R.

Chain rule

Recall the 1-variable chain rule.

or in other notation.

If $y(t) = f(x(t))$ then $y'(t) = f'(x(t))\,x'(t)$

We need a chain rule for the multi-variable setting; i.e. If $y = f(x_1, \ldots, x_n)$ (old) and the x_i in turn are given as functions:

$x_i = u_i(z_1, \ldots, z_n)$ (new) then how does a partial derivative $\frac{\partial y}{\partial z_i}$ depend on all the $\frac{\partial y}{\partial x_j}$ and $\frac{\partial u_i}{\partial z_j}$, for $j = 1, \ldots, n$?

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Derivation in the case of a 2-variable function

Suppose $z = f(x, y)$ and a second coordinate system (s, t) and we can regard x, y as functions of s, t also, so $x = x(s, t)$, $y = y(s, t)$

Start with small increments formula.

$$\Delta z \approx \frac{\partial f}{\partial x} \Delta x + \frac{\partial f}{\partial y} \Delta y$$

and make all these changes relative

to a change Δs, in s.

$$\frac{\Delta z}{\Delta s} \approx \frac{\partial f}{\partial x}\frac{\Delta x}{\Delta s} + \frac{\partial f}{\partial y}\frac{\Delta y}{\Delta s}$$

"dividing across by Δs"

See this now as an ~~equ~~ approximation for the relative change $\frac{\Delta z}{\Delta s}$ as a linear combination of relative changes $\frac{\Delta x}{\Delta s}$, $\frac{\Delta y}{\Delta s}$.

When we take limits here as $\Delta s \to 0$

$$\frac{\Delta z}{\Delta s} \to \frac{\partial z}{\partial s}, \quad \frac{\Delta x}{\Delta s} \to \frac{\partial x}{\partial s}, \quad \frac{\Delta y}{\Delta s} \to \frac{\partial y}{\partial s}$$

and the approximation will become an equality. Which gives us the equation

$$\boxed{\frac{\partial z}{\partial s} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial s}}$$

and similarly

$$\boxed{\frac{\partial z}{\partial t} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial t}}$$

the "chain rule" equations for 2-variables.

and in the multi-variable setting

when $(x_1, \ldots, x_n)$ is replaced by $(s_1, \ldots, s_n)$ we get the set of n-chain rule equations.

$$\frac{\partial z}{\partial s_i} = \sum_{j=1}^{n} \frac{\partial f}{\partial x_j} \frac{\partial x_j}{\partial s_i}, \quad i = 1, \ldots, n$$

From end of section exercises. 2.4.

Q6.

Given $w = x^2 + y^2 + z^2$, the new pair s, t where $x = st$, $y = s\cos t$, $z = s\sin t$

Find $\frac{\partial w}{\partial s}$, $\frac{\partial w}{\partial t}$

Method 1 Direct substitution.

$$w = x^2 + y^2 + z^2$$
$$= (st)^2 + (s\cos t)^2 + (s\sin t)^2$$
$$= s^2t^2 + s^2(\cos^2 t + \sin^2 t)$$
$$= s^2(t^2 + 1)$$

$$\frac{\partial w}{\partial s} = \frac{\partial}{\partial s}\left(s^2(t^2+1)\right) = 2s(t^2+1)$$

$$\frac{\partial w}{\partial t} = \frac{\partial}{\partial t}\left(s^2(t^2+1)\right) = 2s^2t$$

Method 2 Employ the chain rule.

$$\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial s} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial s}$$

$$= 2xt + 2y\cos t + 2z\sin t$$

$$= 2st^2 + 2s\cos^2 t + 2s\sin^2 t$$

$$= 2st^2 + 2s$$

$$= 2s(t^2+1), \text{ as expected from method 1.}$$

And

$$\frac{\partial w}{\partial t} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial t} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial t}$$

$$= 2xs + 2y(-s\sin t) + 2z\, s\cos t.$$

$$= 2s^2 t - 2s^2\cos t\sin t + 2s^2\sin t\cos t.$$

$$= 2s^2 t, \text{ as expected from before.}$$

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23 (a) Given the differentiable single variable function f. Consider $w(x,y)$

$$w(x,y) = e^{-y} f(x-y)$$

Prove this is a solution to

$$w + \frac{\partial w}{\partial x} + \frac{\partial w}{\partial y} = 0.$$

Focussing on f, f has a derivative $\frac{df}{dt}$, and now $t = x - y$

so $\frac{\partial f}{\partial x} = \frac{df}{dt}\frac{\partial t}{\partial x} = \frac{df}{dt}$

& $\frac{\partial f}{\partial y} = \frac{df}{dt}\frac{\partial t}{\partial y} = -\frac{df}{dt}$

And now

$$\frac{\partial w}{\partial x} = \frac{\partial}{\partial x}\left(e^{-y} f(x-y)\right)$$

$$= e^{-y}\frac{df}{dt}, \quad (t = x-y)$$

$$\frac{\partial w}{\partial y} = \frac{\partial}{\partial y}\left(e^{-y} f(x-y)\right)$$

$$= e^{-y}\frac{\partial}{\partial y} f(x-y) + \frac{\partial}{\partial y}(e^{-y}) f(x-y)$$

$$= e^{-y}\left(-\frac{df}{dt}\right) - e^{-y} f(x-y).$$

So Looking at the PDE

$$w + \frac{\partial w}{\partial x} + \frac{\partial w}{\partial y}$$

$$= e^{-y} f(x-y) + e^{-y} \frac{df}{dt} - e^{-y} \frac{df}{dt}$$

$$- e^{-y} f(x-y)$$

$$= 0.$$

So this w is a solution of the PDE.

Look again at the chain rule eqns. $(x_1, \dots, x_n) \longrightarrow (s_1, \dots, s_n)$.

$$\frac{\partial z}{\partial s_i} = \sum_{j=1}^{n} \frac{\partial f}{\partial x_j} \frac{\partial x_j}{\partial s_i}$$

Stare at this and think of linear algebra

$$\square_i = \sum_{j=1}^{n} \square_j \square_{j,i}$$

vector vector matrix

this could show a row vector as a product of a row vector multiplied

by a matrix.

In 2-variable case this would be.

$$\left(\frac{\partial z}{\partial s}, \frac{\partial z}{\partial t}\right) = \left(\frac{\partial z}{\partial x}, \frac{\partial z}{\partial t}\right) \begin{pmatrix} \frac{\partial x}{\partial s} & \frac{\partial x}{\partial t} \\ \frac{\partial y}{\partial s} & \frac{\partial y}{\partial t} \end{pmatrix}$$

this is called the Jacobian matrix of the coordinate transformation

$$(x, y) \longrightarrow (s, t).$$

In general "the Jacobian" refers to the determinant of this matrix, which will play an important role in certain multivariable integrals.

$$\frac{\partial(x, y)}{\partial(s, t)} = \begin{vmatrix} \frac{\partial x}{\partial s} & \frac{\partial x}{\partial t} \\ \frac{\partial y}{\partial s} & \frac{\partial y}{\partial t} \end{vmatrix}$$

In general.

$$\frac{\partial(x_1, \ldots, x_n)}{\partial(s_1, \ldots, s_n)} = \left| \left(\frac{\partial x_i}{\partial s_j} \right)_{i,j} \right|$$

$n \times n$ matrix of those partial derivatives.